- IPv6 address is a 128 bit unique address given to a computer or a device. It follows Hexadecimal number notation.
- URL (Uniform Resource Locator) is the address of a document on the Internet. URL is divided into two types: Absolute URL and Relative URL
- URL is made up of four parts- protocols, hostname, folder name and file name. Absolute URL contains all the four necessary and fundamental parts of URL.
- Relative URL contains only folder name and the file name or just the file name.
- There are 3 important components in the Domain Name System. They are **Namespace, Name server and Zone.**
- **Label** is a string which can have maximum of 63 characters. Each node in that level should have unique label.
- Domain name space is a tree like structure with root element on the top. It can have a maximum of 128 levels starting from root element taking the level 0 to level 127.
- Domain name is the sequence of labels separated by dot (.). The domain name is always read from the leaf node to root node. The root node always represent NULL string. So All the domain name ends with dot.
- In the domain name space (DNS) tree structure **domain** is a sub structure tree. The domain can be further divided into sub domains.
- Name Servers are programs that run on a physical system and store all the zone data. It provides to clients when queried by them.
- Zone is the contiguous part up to which the server has access. The domain assigned for the server does not divide into further sub domains then zone is same as domain.
- Resolver, a client/ server application, initiates the process of resolving the domain names.

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